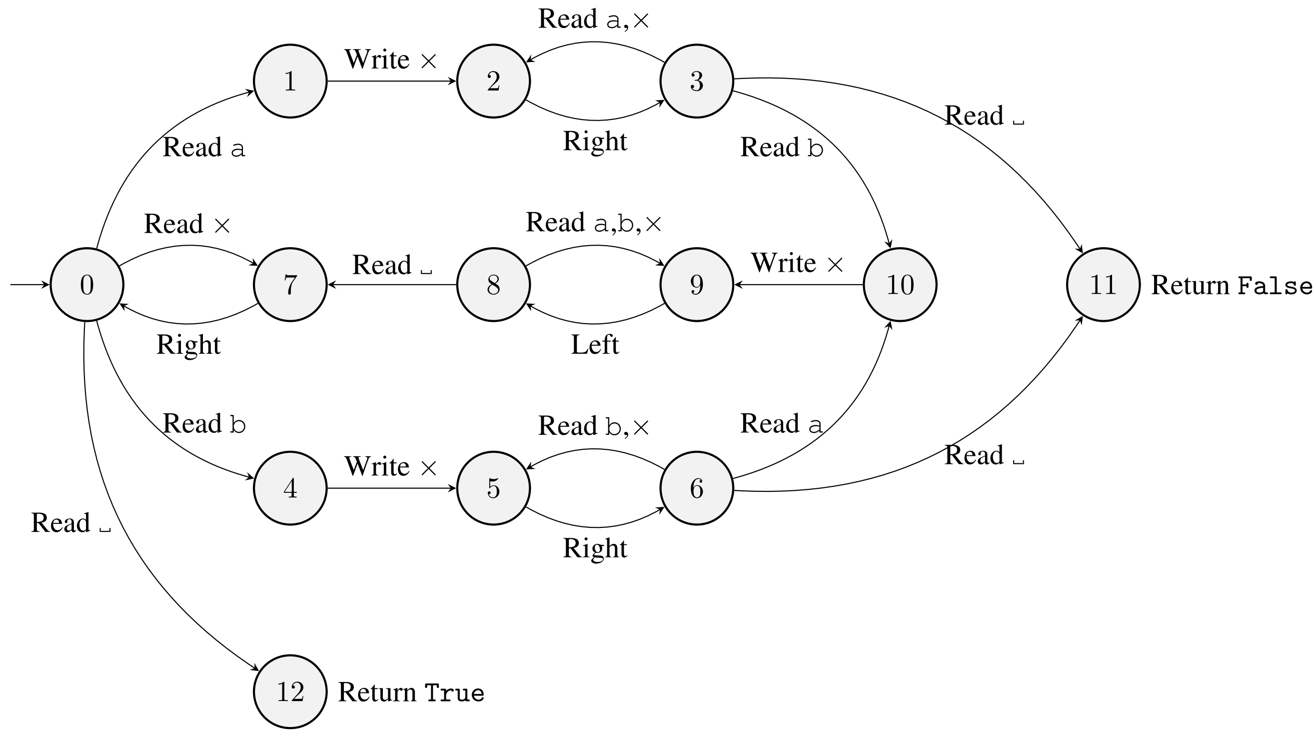
Theories of Computation: Summative Assignment 2

To be handed in on Canvas before **Tuesday 21st March, 12pm**

Consider the deterministic Turing machine $\mathcal{M}$ on alphabet $\{\mathtt{a}, \mathtt{b}, \times, ␣\}$ with return value set $V = \{\mathtt{True}, \mathtt{False}\}$. The tape initially contains a non-empty block of a's and b's on an otherwise blank tape, with the head on the leftmost non-blank character. The transition function is given by the diagram below



Questions.

1. (a) Give the complete run of machine $\mathcal{M}$ on the word aa, with the head positioned on the first a.

   That is, the starting confiiguration of the tape is: ␣ a a ␣ (head • on the first a)

   At each step, indicate the tape contents, the position of the head, the current state and the instruction (including the result if it is a Read). **[2 marks]**

   (b) Give the first seven steps of the run of machine $\mathcal{M}$ on the word ab, with the head positioned on the a.

   That is, the starting confiiguration of the tape is: ␣ a b ␣ (head • on the a)

   At each step, indicate the tape contents, the position of the head, the current state and the instruction (including the result if it is a Read). **[2 marks]**

2. (a) Is abaa in the language of $\mathcal{M}$? **[1 mark]**

   (b) Is abba in the language of $\mathcal{M}$? **[1 mark]**

   (c) What is the language of $\mathcal{M}$? **[2 marks]**

3. For $w \in \{\mathtt{a}, \mathtt{b}\}^*$, let $T(w)$ denote the processing time of machine $\mathcal{M}$ on word w.

   (a) Give a word w_n of length $n > 0$ such that $T(w_n)$ is linear in n. Justify your answer by giving $T(w_n)$ as a function of n and showing that it is in $\mathcal{O}(n)$. **[3 marks]**

   (b) Give a word v_n of length $n > 0$ (you may assume n is even) such that $T(v_n)$ is quadratic in n. Justify your answer by giving $T(v_n)$ as a function of n and showing that it is in $\mathcal{O}(n^2)$. **[3 marks]**

4. (a) Construct a deterministic Turing machine $\mathcal{N}$ on alphabet $\{\mathtt{a}, \mathtt{b}, \llcorner\!\lrcorner\}$ with **seven** states that decides the language described by regular expression $\mathtt{aa}^*\mathtt{bb}^*$.

   Assume that the tape initially contains a non-empty block of a's and b's on an otherwise blank tape, with the head on the **rightmost** non-blank character.

   The position of the head at the end of the run does not matter. **[4 marks]**

   (b) Explain briefly how you would use machines $\mathcal{M}$ and $\mathcal{N}$ as macros to construct a Turing machine that recognizes language $\mathcal{L} = \{a^n b^n \mid n \geq 1\}$.

   That is, a machine that starts on a tape that initially contains a non-empty block of a's and b' on an otherwise blank tape (which represents a word $w \in \{\mathtt{a}, \mathtt{b}\}^*$) with the head on the **leftmost** non-blank character and returns True when $w \in \mathcal{L}$ and returns False when $w \notin \mathcal{L}$.

   The position of the head at the end of the run does not matter. **[2 marks]**